

- (b) $=0.007184 \times B^2 \times 0.425 \times C^{0.725}$ (c) See table above
 (d) For the given group the Mosteller formula gives a slightly larger BSA. There is one obvious exception that being Eva who is very much underweight for her height (average female height is 162.1 cm and weight 71 kg)
 12. (a) and (b) Adam 2181.3, 3763; Bella 1392.02, 1914
 Cathy 1571.58, 2436; David 2210.06, 2652; Eva 1342.32, 1611; Frank 2380.86, 3274; Gary 1762.86, 3041; Heath 2221.64, 4221; Ian 2081.64, 3227

CHAPTER 9 REVIEW EXERCISE

1. (a) 28 (b) 5 2. (a) 28 (b) 75 3. (a) $5\frac{1}{3}$ (b) 21 (c) 6 (d) -9
 4. (a) 63 (b) $1\frac{4}{9}$ 5. (a) 24 (b) 849 6. (a) 17.3 m (b) 46 cm
 (c) 56 cm (d) 1.75 mm 7. (a) 2.8 (b) 40.6 (c) -6 (d) 17
 8. (a) 20 km (b) 5 km (c) Approx. 128 m above the Earth's surface. 9. (a) 6 (b) 3 10. (a) $A = 2.4$, $B = 4.2$, $C = 4.8$, $D = 4.2$ (b) (iii), (iv) and (vi) (c) 54 11. (a) 160.8 m (b) 6.4 m
 12. (a) 1.9775m^2 (4 d.p.) (b) 2.1934m^2 (4 d.p.)
 13. (a) $(i) = B^3 + 1$ (ii) $= C^2 + 2$ (iii) $= \pi \times B^3 \times 2 \times C^2$
 (b) 452.39cm^3 (c) 1847.3cm^3 (d) 4 cm (e) 6 cm
 (f) 1579.63cm^3 (g) 19950.37cm^3

EXERCISE 10A

1. Matrix A (a) square matrix, (b) 3×3 ;
 Matrix B (a) square matrix and an identity matrix, (b) 2×2 ;
 Matrix C (a) row matrix, (b) 1×4 ;
 Matrix D (a) square matrix and a zero matrix, (b) 4×4 ;
 Matrix E (a) none of these, (b) 2×4 ;
 Matrix F (a) column matrix, (b) 3×1 ;
 Matrix G (a) none of these, (b) 2×3 ;
 Matrix H (a) square matrix and an identity matrix, (b) 3×3 .
 2. (a) 1×2 (b) 2×3 (c) 3×4 (d) 3×3 (e) 4×2 (f) 4×4
 (g) 1×5 (h) 1×1
 3. (a) $\begin{bmatrix} 7 \\ 12 \end{bmatrix}$ (b) $\begin{bmatrix} 8 & 0 \\ 0 & 6 \end{bmatrix}$ (c) $\begin{bmatrix} -2 \\ -4 \\ -6 \end{bmatrix}$ (d) $\begin{bmatrix} 2 & 3 & -3 \\ 3 & -4 & 5 \end{bmatrix}$ (e) not possible
 4. (a) $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$ (b) $\begin{bmatrix} -6 & 4 \\ -2 & 12 \end{bmatrix}$ (c) $\begin{bmatrix} 5 \\ -6 \\ 8 \end{bmatrix}$ (d) $\begin{bmatrix} -3 & 1 \\ 4 & 0 \\ -3 & -2 \end{bmatrix}$ (e) $\begin{bmatrix} 4 & 2 & -5 \\ 3 & 1 & 1 \\ -2 & 5 & 3 \end{bmatrix}$
 5. (a) $\begin{bmatrix} 7 & -1 & 9 \\ 1 & 4 & 7 \end{bmatrix}$ (b) $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ (c) $\begin{bmatrix} 0 & -5 \\ 0 & 0 \\ -2 & -4 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 6 \\ -3 & -5 \\ 2 & -11 \end{bmatrix}$
 6. Different order matrices cannot be added.
 7. (a) 8 (b) 1 (c) -4 (d) 0 (e) 4 (f) 20 (g) -3 (h) -11
 (i) $i = 2, j = 3$ (j) $i = 3, j = 2$ (k) $i = 1, j = 2$ (l) $i = 2, j = 1$.
 8. (a) (i) $\begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$ (ii) $\begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$ (iii) Addition of matrices is commutative (b) (i) $\begin{bmatrix} 3 & -6 \\ 5 & 1 \end{bmatrix}$ (ii) $\begin{bmatrix} -3 & 6 \\ -5 & -1 \end{bmatrix}$ (iii) Subtraction of matrices is not commutative.
 9. (a) $x = 3$ (b) $x = 2$, $y = -3$ (c) $x = 3.5$, $y = -3$ (d) $x = -1$, $y = 3$.
 10. (a) $\begin{bmatrix} 6 \\ 10 \end{bmatrix}$ (b) $\begin{bmatrix} 6 & 3 & 12 \\ 0 & -6 & -9 \\ 0 & 0 & 6 \end{bmatrix}$ (c) $\begin{bmatrix} 4 \\ -3 \\ 0 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & -2 \\ -0.6 & 0.2 \end{bmatrix}$ (e) $\begin{bmatrix} -4 \\ 3 \\ -2 \end{bmatrix}$

11. (a) $A + C = \begin{bmatrix} 1 & 8 \\ 3 & 3 \end{bmatrix}$ (b) $A + D = \begin{bmatrix} -19 \\ 2 \\ -5 \end{bmatrix}$ 12. (a) $\begin{bmatrix} 5 & -5 \\ 6 & -1 \end{bmatrix}$
 (b) $\begin{bmatrix} 4 \\ -7 \\ 9 \end{bmatrix}$ (c) $\begin{bmatrix} 5 & -1 \\ -0.5 & -2.25 \\ 2.5 & -3.5 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 0 & 0 \\ -1 & -1 & -2 \end{bmatrix}$ (e) $\begin{bmatrix} 3.5 & 7.4 \\ 10.6 & -7.2 \end{bmatrix}$

13. (a) $\begin{bmatrix} -17 & -19 & -23 \\ 49 & 4 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} -11 & 29 & -6 \\ 30 & -53 & -30 \end{bmatrix}$
 (c) $\begin{bmatrix} -1 & -125 & -28 \\ 8 & 167 & 92 \end{bmatrix}$ (d) $\begin{bmatrix} 9 & 75 & 6 \\ 123 & -204 & -159 \end{bmatrix}$
 (e) $\begin{bmatrix} 160 & -2 & 144 \\ -201 & 15 & -59 \end{bmatrix}$ (f) $\begin{bmatrix} -96 & -70 & -95 \\ 73 & 125 & 125 \end{bmatrix}$

14. (a) $w = -6$; $x = -2$; $y = 2$; $z = 4$ (b) $w = 4$; $x = 3$; $y = -4$; $z = 0$ (c) $w = 4$; $x = -6$; $y = -4$; $z = -2$ (d) $w = 1$; $x = -4$; $y = 3$; $z = \pm 3$ (e) $w = -4$; $x = -5$; $y = 1$; $z = 0.25$

15. (a) (i) 4 (ii) 1 (iii) 0 (b) (i) $i = 1$; $j = 3$ (ii) $i = 3$; $j = 1$ (iii) $i = 3$; $j = 2$ (c) (i) 3 (ii) 8 (iii) 9

16. (a) (i) 1 (ii) -2 (iii) 15 (b) (i) 3 (ii) 0 (iii) 1 (c) 4 17. $k = 2$

18. $2(A - B) = 2 \left(\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & 1 \end{bmatrix} - \begin{bmatrix} -1 & 1 & 0 \\ 4 & 3 & -2 \end{bmatrix} \right)$
 $= 2 \left(\begin{bmatrix} 2 & -2 & 2 \\ -4 & -1 & 3 \end{bmatrix} \right) = \begin{bmatrix} 4 & -4 & 4 \\ -8 & -2 & 6 \end{bmatrix}$
 $2A - 2B = 2 \left(\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & 1 \end{bmatrix} \right) - 2 \left(\begin{bmatrix} -1 & 1 & 0 \\ 4 & 3 & -2 \end{bmatrix} \right)$
 $= \begin{bmatrix} 2 & -2 & 4 \\ 0 & 4 & 2 \end{bmatrix} - \begin{bmatrix} -2 & 2 & 0 \\ 8 & 6 & -4 \end{bmatrix} = \begin{bmatrix} 4 & -4 & 4 \\ -8 & -2 & 6 \end{bmatrix}$
 $= 2(A - B)$

19. (a) Not possible, matrices cannot be added as they are not

of the same order. (b) -9 (c) $\begin{bmatrix} -2 & 5 \\ -2 & -6 \end{bmatrix}$ (d) $\begin{bmatrix} 2 & 3 \\ -6 & 2 \end{bmatrix}$

(e) $\begin{bmatrix} 0 & 0 & -2 \\ 2 & 1 & -1 \\ -2 & -3 & 2 \end{bmatrix}$ (f) Not possible, identity matrix for a 2×3 matrix does not exist.

(g) $\begin{bmatrix} 2 & -2 & 7 \\ 1 & 2 & -3 \end{bmatrix} + \begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix}$ (h) $\begin{bmatrix} -1 & -6 \\ 0 & 5 \end{bmatrix}$

20. $\begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}$ 21. $\begin{bmatrix} 3 & -1 & -2 \\ -3 & 5 & -4 \\ -1 & 3 & -3 \end{bmatrix}$

22. (a) $F = \begin{bmatrix} \text{Milk} & \text{Dark} \\ \text{Hard} & 8 & 5 \\ \text{Soft} & 9 & 6 \end{bmatrix}$ (b) 15, there are 15 milk chocolate in the Roses box

(c) $J = 2R + 3F$ $J = \begin{bmatrix} 42 & 25 \\ 39 & 26 \end{bmatrix}$ (d) 132 chocolates

23. (a) $\begin{bmatrix} 62000 & 113000 \\ 103000 & 147000 \\ 19000 & 24000 \end{bmatrix}$ Total costs for both sites.

(b) $\begin{bmatrix} 31000 & 56500 \\ 51500 & 73500 \\ 9500 & 12000 \end{bmatrix}$ Average costings over the two sites.

(c) $\begin{bmatrix} 2000 & 3000 \\ 3000 & 7000 \\ 1000 & 2000 \end{bmatrix}$ Extra costs involved for developing site B as compared to site A.

24. (a) $\begin{bmatrix} 3 & 12 & 5 \\ 2 & 17 & 4 \\ 3 & 16 & 2 \\ 4 & 16 & 5 \end{bmatrix}$ Total sales for both stores of each musical instrument week by week for January 2014.

(b) $\begin{bmatrix} 200 & 800 & 400 \\ 100 & 1100 & 200 \\ 300 & 900 & 100 \\ 200 & 700 & 300 \end{bmatrix}$ (c) $\begin{bmatrix} 290 & 1160 & 490 \\ 190 & 1640 & 380 \\ 300 & 1530 & 190 \\ 380 & 1510 & 480 \end{bmatrix}$

(d) P shows the weekly profit of both stores over the 4 weeks for each instrument. (e) \$8540